Cybersecurity Awareness and Incident Response Among Small Businesses in Orlando

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Proposed Start Date: January 1, 2027

Background

In Orlando, many small businesses operate across tourism, hospitality, and retail sectors and are known for handling sensitive customer payment and personal data. This makes them attractive targets for cybercriminals. Despite the scale of the threat, cybersecurity preparedness of these businesses in Central Florida remains limited.

This project will survey these owners to gather data and understand how much they know about common cyber threats, whether they have basic protective measures in place, and what they would actually do if something went wrong. The goal is to find out where the gaps are so that targeted education can fill them.

Methodology

Research Design

This study will use a descriptive survey to gather data from these businesses. The survey will be built around the NIST Cybersecurity Framework's five core functions , Identify, Protect, Detect, Respond, and Recover.It will be written in plain language so it's accessible to business owners who aren't familiar with technology can understand (NIST, 2018).

Survey questions will cover three main areas:

- Threat familiarity , whether respondents can recognize phishing attempts, understand what ransomware is, and identify social engineering tactics
- Protective measures , whether their business uses basic tools like multi-factor authentication, password managers, or regular software updates
- Incident response , whether their business has any kind of plan for what to do after a breach, even an informal one

Data Collection

Surveys will be distributed digitally through the UCF research participant pool and the Orlando Small Business Development Center (SBDC) network. I will personally reach out to the SBDC and the Orlando Chamber of Commerce to distribute the survey link through their member email lists, which include hundreds of local small businesses across the hospitality, retail, and tourism sectors. Flyers will also be handed out at UCF small businesses showcase and personally to local shop owners.

A target sample of 75 to 100 completed responses is realistic given the distribution channels and the 10-week collection window. The survey will be hosted using UCF's existing Qualtrics license at no additional cost.

Analysis

Once the surveys are collected, I'll look at the numbers to spot patterns, like whether newer businesses tend to be less prepared, or whether certain industries know more about cybersecurity than others. Any written responses will be grouped by common themes to round out what the numbers show.

Everything will be wrapped up into a short written summary and a simple infographic that local business organizations and the SBDC can hand out and use.

Outcomes

This project will produce:

- A dataset on cybersecurity awareness and incident response, which can be used as a baseline for future research
- A written research report summarizing findings and key themes
- A poster presentation at the UCF SURE Conference

References

Alahmari, A., & Duncan, B. (2020). Cybersecurity risk management in small and medium-sized enterprises: A systematic review of recent evidence. 2020 International Conference on Cyber Situational Awareness, Data Analytics and Assessment. IEEE. https://doi.org/10.1109/CyberSA49311.2020.9139638

National Institute of Standards and Technology. (2018). Framework for improving critical infrastructure cybersecurity (Version 1.1). U.S. Department of Commerce. https://doi.org/10.6028/NIST.CSWP.04162018

United States Small Business Administration. (2023). Small business cybersecurity. SBA.gov. https://www.sba.gov/business-guide/manage-your-business/small-business-cybersecurity

Verizon. (2023). 2023 data breach investigations report. Verizon Business. https://www.verizon.com/business/resources/reports/dbir/

Project Timeline

Start Date: January 1, 2027 - End Date: April 30, 2027

Dates	Tasks
Jan 1 to Jan 31, 2027	Conduct literature review on small business cybersecurity; draft and refine survey questions in Qualtrics; identify and contact SBDC and Chamber of Commerce distribution contacts
Feb 1 to Feb 28, 2027	Finalize survey; confirm event distribution plan; begin outreach to SBDC and local businesses; attend UCF Small Business Showcase to distribute survey link
Mar 1 to Mar 31, 2027	Continue active survey distribution; visit small businesses in the area if need more data
Apr 1 to Apr 14, 2027	Close survey collection; clean and organize dataset; run descriptive statistical analysis; begin thematic coding of open-ended responses; attend SBDC workshop to distribute survey as needed
Apr 15 to Apr 30, 2027	Complete analysis; draft written report and public-facing infographic; review with faculty mentor; finalize SURE conference poster and submit grant completion materials; present findings

Budget

Total Requested: $290

Item	Description / Justification	Cost
Participant Incentives	Small raffle incentive (e.g., $25 gift cards x 8) to encourage survey completion	$200
Flyer Printing	Printed QR code flyers for in-person distribution at UCF Small Business Showcase and local businesses, approximately 90 copies at UCF Print Services rates	$45
Supplies	Notebooks and folders for literature review notes, data organization, and meeting materials throughout the project	$45
Total		**$290**